

MATHEMATICAL TOURNAMENTS: RÉDEI'S THEOREM AND LANDAU'S THEOREM

MATH 38: GRAPH THEORY

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1 Introduction

Graph theory provides powerful tools for modeling real-world structures, and one application that caught our eye is **tournaments**. Inherently, it is an interesting idea to consider relative comparisons and wins within a graph. How do you structure a win or loss? Can the definition of a tournament go beyond our colloquial use within sports?

When expressing the distinction between two nodes, it is natural to consider directed graphs, since the arrow creates a binary outcome. Hence, we introduce *tournament* as an orientation of a complete graph. For every pair of distinct vertices u, v , exactly one of the directed edges $u \rightarrow v$ or $v \rightarrow u$ is present. Despite this innocuous definition, tournaments exhibit deep structural rigidity that has fascinated combinatorialists for nearly a century.

Tournaments are necessary in areas of combinatorics, order theory, and applied mathematics. The orientation of each individual edge is unconstrained, yet two classical theorems demonstrate that the global structure is far more disciplined than one might expect:

- **Rédei's Theorem** (1934): every tournament contains a Hamiltonian path, providing a “ranking” of all participants.
- **Landau's Theorem** (1953): a nondecreasing sequence of nonnegative integers is realizable as a score sequence of a tournament *if and only if* it satisfies a clean family of inequalities.

In this final report we develop the necessary background, present proofs of both theorems, and work through several examples.

2 Background

Definition 2.1. Directed graph

A *directed graph* (or *digraph*) is a pair $D = (V, E)$ where V is a finite set of *vertices* and $E \subseteq \{(u, v) \in V \times V : u \neq v\}$ is a set of ordered pairs called *directed edges* (or *arcs*). We write $u \rightarrow v$ for the arc (u, v) .

Definition 2.2. Tournament, Outdegree, and score sequence

A *tournament* on n vertices is a directed graph $T = (V, E)$ such that for every pair of distinct vertices $u, v \in V$, exactly one of the arcs $u \rightarrow v$ or $v \rightarrow u$ is present. The *outdegree* of a vertex v in a tournament T is

$$d^+(v) = |\{u \in V : v \rightarrow u\}|.$$

Listing the outdegrees in nondecreasing order $d_1 \leq d_2 \leq \dots \leq d_n$ yields the *score sequence* of T .

Definition 2.3. Hamiltonian path

A *Hamiltonian path* in a digraph is a directed path that visits every vertex exactly once.

Definition 2.4. Transitive tournament

A tournament T is *transitive* if $u \rightarrow v$ and $v \rightarrow w$ together imply $u \rightarrow w$. Equivalently, T is transitive iff it is acyclic, iff its arcs encode a linear order on V .

Proposition 2.1. Score sum

The score sequence of any tournament on n vertices sums to $\binom{n}{2}$.

Proof. A tournament on n vertices has exactly $\binom{n}{2}$ arcs, since this is the number of unordered pairs of distinct vertices and each contributes exactly one arc. Each arc $v \rightarrow u$ contributes exactly 1 to the outdegree of its tail v . Summing over all arcs gives

$$\sum_{v \in V} d^+(v) = |E(T)| = \binom{n}{2}. \quad \square$$

3 Rédei's Theorem

Theorem 3.1. Rédei, 1934

Every tournament contains a Hamiltonian path.

Proof. We proceed by induction on the number of vertices n .

Base case. If $n = 1$, the single vertex itself is a Hamiltonian path (of length 0).

Inductive step. Assume every tournament on $n - 1$ vertices contains a Hamiltonian path. Let T be a tournament on n vertices, pick any vertex $v \in V(T)$, and consider the subtournament $T - v$ on the remaining $n - 1$ vertices. By induction there is a Hamiltonian path of $T - v$,

$$P : v_1 \rightarrow v_2 \rightarrow \dots \rightarrow v_{n-1}.$$

We now insert v into P to obtain a Hamiltonian path of T . There are three cases.

Case 1: $v \rightarrow v_1$. Insert v to the front of P to obtain the Hamiltonian path $v \rightarrow v_1 \rightarrow \dots \rightarrow v_{n-1}$.

Case 2: $v_{n-1} \rightarrow v$. Insert v to the end of P to obtain $v_1 \rightarrow \dots \rightarrow v_{n-1} \rightarrow v$.

Case 3: Neither of the two cases. Since T is a tournament, exactly one of $v_1 \rightarrow v$ or $v \rightarrow v_1$ holds; the assumption of Case 3 rules out the latter, so $v_1 \rightarrow v$. Similarly $v \rightarrow v_{n-1}$. Now look at the arcs between v and each v_i as i ranges from 1 to $n-1$. At $i=1$ the arc points toward v ($v_1 \rightarrow v$), and at $i=n-1$ it points away from v ($v \rightarrow v_{n-1}$). Hence there must be some smallest index i where the direction switches: $v_i \rightarrow v$ but $v \rightarrow v_{i+1}$. Inserting v between v_i and v_{i+1} yields the Hamiltonian path

$$v_1 \rightarrow \cdots \rightarrow v_i \rightarrow v \rightarrow v_{i+1} \rightarrow \cdots \rightarrow v_{n-1}.$$

□

4 Landau's Theorem

Theorem 4.1. Landau, 1953

A nondecreasing sequence of nonnegative integers $d_1 \leq \cdots \leq d_n$ is the score sequence of some tournament on n vertices if and only if

$$\sum_{i=1}^k d_i \geq \binom{k}{2} \quad \text{for all } 1 \leq k \leq n,$$

with equality at $k=n$.

4.1 Interpretation

Any k vertices span $\binom{k}{2}$ internal arcs, each a win for someone inside the set; outside wins only add more. So their outdegrees sum to at least $\binom{k}{2}$, tightest when the k are the *weakest* players. Equality at $k=n$ is forced by Proposition 2.1.

4.2 Necessity Proof

Label vertices so $d^+(v_i) = d_i$, fix k , and set $S = \{v_1, \dots, v_k\}$. The subtournament $T[S]$ has $\binom{k}{2}$ arcs, so $\sum_{v \in S} d_{T[S]}^+(v) = \binom{k}{2}$. Since $d_T^+(v) \geq d_{T[S]}^+(v)$ for each $v \in S$,

$$\sum_{i=1}^k d_i = \sum_{v \in S} d_T^+(v) \geq \binom{k}{2},$$

with equality at $k=n$ by Proposition 2.1.

4.3 Sufficiency Proof

The classical sufficiency proof is combinatorial: take a counterexample minimizing $\sum_i d_i^2$ and reverse one arc to reach a contradiction (the argument we sketched in our presentation; see West [2] or Bondy–Murty [1]). However, in this report, we instead give a proof by network flows, which reuses the max-flow–min-cut machinery from Chapter 4 (and our lectures) and reduces realizability to a max-flow problem as shown:

$$\underbrace{\text{tournament with scores } d \text{ exists}}_{\text{goal}} \xLeftrightarrow{(A)} \text{ flow of value } \binom{n}{2} \text{ in } N \xLeftrightarrow{(B)} \text{ every cut } \geq \binom{n}{2} \xLeftrightarrow{(C)} \text{ Landau holds.}$$

Step (A): Build the network N . Source s , sink t ; a game node g_{ij} for each pair $i < j$; a player node p_i for each vertex. Arcs: $s \rightarrow g_{ij}$ with capacity 1; $g_{ij} \rightarrow p_i$ and $g_{ij} \rightarrow p_j$ with capacity 1; $p_i \rightarrow t$ with capacity d_i .

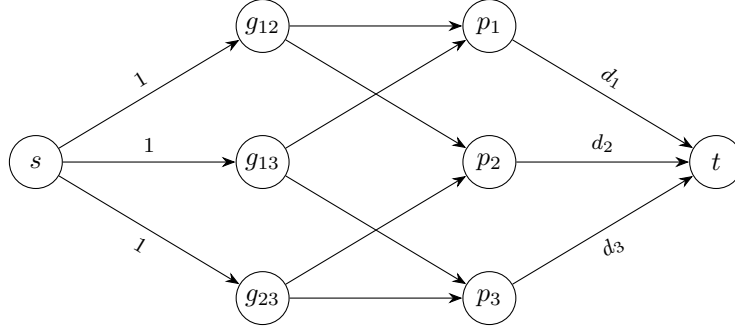


Figure 1: The network N for $n = 3$. An integral flow of value $\binom{3}{2} = 3$ assigns each game a winner and realizes (d_1, d_2, d_3) .

An integral flow of value $\binom{n}{2}$ saturates every $s \rightarrow g_{ij}$; by integrality, each game node forwards its unit to exactly one player, orienting that game. Conversely, any tournament with scores d defines such a flow by routing each game through its winner.

Step (B): Max-flow min-cut. Since the source arcs cap the flow at $\binom{n}{2}$, by the theorem a flow of value $\binom{n}{2}$ exists *iff* every s - t cut has capacity at least $\binom{n}{2}$.

Step (C): Cut computation. Let $A \subseteq \{1, \dots, n\}$ index the player nodes on the source side. The cut severs each $p_i \rightarrow t$ for $i \in A$ (contributing $\sum_{i \in A} d_i$) and each $s \rightarrow g_{ij}$ where $\{i, j\} \not\subseteq A$ (contributing $\binom{n}{2} - \binom{|A|}{2}$). The total capacity is

$$\binom{n}{2} - \binom{|A|}{2} + \sum_{i \in A} d_i \geq \binom{n}{2} \iff \sum_{i \in A} d_i \geq \binom{|A|}{2}.$$

For $|A| = k$ this is tightest when $A = \{1, \dots, k\}$, recovering exactly the Landau inequalities. $\square$

4.4 A structural dividend: strictness and connectivity

The strict-vs-equality distinction is meaningful. If $\sum_{i=1}^k d_i = \binom{k}{2}$ for some $k < n$, the k weakest players win only among themselves and lose every outside game; no arc leaves that set, so the tournament is not strongly connected. Strict inequality everywhere forces each bottom set to win outward, exactly enough to glue it together (Moon [5]).

Theorem 4.2. Moon's connectivity criterion

A tournament with sorted score sequence $d_1 \leq \dots \leq d_n$ is strongly connected *iff* $\sum_{i=1}^k d_i > \binom{k}{2}$ for every $1 \leq k \leq n-1$.

Remark 4.1. Our examples through Moon's lens

The cyclic triangle $(1, 1, 1)$ is strict throughout ($1 > 0$, $2 > 1$), so strongly connected. The transitive tournament $(0, 1, 2)$ hits equality at $k = 1, 2$, correctly flagging it as acyclic.

5 Examples

Example 5.1. Three vertices

Up to isomorphism there are exactly two tournaments on three vertices.

- The *transitive* tournament with arcs $a \rightarrow b$, $b \rightarrow c$, $a \rightarrow c$ has score sequence $(0, 1, 2)$ and the unique Hamiltonian path $a \rightarrow b \rightarrow c$.
- The *cyclic* tournament with arcs $a \rightarrow b$, $b \rightarrow c$, $c \rightarrow a$ has score sequence $(1, 1, 1)$ and *three* Hamiltonian paths: $a \rightarrow b \rightarrow c$, $b \rightarrow c \rightarrow a$, and $c \rightarrow a \rightarrow b$. This could be used to model Rock, Paper, and Scissors

Both score sequences satisfy Landau's inequalities. For $(1, 1, 1)$,

$$k = 1 : 1 \geq 0; \quad k = 2 : 1 + 1 = 2 \geq 1; \quad k = 3 : 1 + 1 + 1 = 3 = \binom{3}{2}.$$

Example 5.2. A non-realizable sequence

Consider the candidate sequence $(0, 0, 2, 2)$ for $n = 4$. The sum is $4 = \binom{4}{2}$, which looks promising, but at $k = 2$ we have $0 + 0 = 0 < 1 = \binom{2}{2}$. Landau's condition fails, so this is *not* a score sequence. Intuitively, two vertices with outdegree 0 would each have to lose every game—but they play each other, so one of them must win at least once.

Example 5.3. A realizable sequence

The sequence $(1, 1, 2, 2)$ has sum $6 = \binom{4}{2}$ and satisfies Landau's theorem because

$$k = 1 : 1 \geq 0; \quad k = 2 : 2 \geq 1; \quad k = 3 : 4 \geq 3; \quad k = 4 : 6 = 6.$$

A tournament realizing this sequence on vertices $\{a, b, c, d\}$ has arcs

$$a \rightarrow b, \quad b \rightarrow c, \quad b \rightarrow d, \quad c \rightarrow a, \quad c \rightarrow d, \quad d \rightarrow a.$$

We check: $d^+(a) = 1$ (beats b), $d^+(b) = 2$ (beats c, d), $d^+(c) = 2$ (beats a, d), $d^+(d) = 1$ (beats a). Sorted, the outdegrees are $(1, 1, 2, 2)$. A Hamiltonian path is $a \rightarrow b \rightarrow c \rightarrow d$, as guaranteed by Rédei's theorem.

6 Conclusion

Rédei's and Landau's theorems together show the rigidity of tournament structure. The first guarantees that there is always a linear ranking via a Hamiltonian path. The second tells us exactly which profiles of wins can occur. More importantly, these theorems build the framework for topics such as tournaments, kings and kernels, transitive subtournaments, and probabilistic tournament theory.

References

- [1] J.A. Bondy and U.S.R. Murty, *Graph Theory*, Graduate Texts in Mathematics 244, Springer, 2008.
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- [5] J.W. Moon, *Topics on Tournaments*, Holt, Rinehart and Winston, 1968.